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QUARTERLY EXAMINATION -2026

CLASS: 12 **MATHEMATICS** **Reg.No**

Time : 3.00 Hours **MARKS : 90**

PART – I

I. Answer all the questions . **20 x 1 = 20**

1. If A, B and C are invertible matrices of some order, then which one of the following is not true?
   1) adj A = |A| A^{-1} 2) adj (AB) = (adj A) (adj B)
   3) det A^{-1} = (det A)$^{-1}$ 4) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

2. If $A = \begin{bmatrix} 1 & \tan\frac{\theta}{2} \\ -\tan\frac{\theta}{2} & 1 \end{bmatrix}$ and $AB = I_2$, then B =
   1) $\left(\cos^2\frac{\theta}{2}\right)A$ 2) $\left(\cos^2\frac{\theta}{2}\right)A^T$ 3) $(\cos^2\theta)I$ 4) $\left(\sin^2\frac{\theta}{2}\right)A$

3. The augmented matrix of a system of linear equations is $\begin{bmatrix} 1 & 2 & 7 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & \lambda-7 & \mu+5 \end{bmatrix}$. The system has infinitelymany solutions if
   1) $\lambda = 7, \mu \neq -5$ 2) $\lambda = -7, \mu = 5$ 3) $\lambda \neq 7, \mu \neq -5$ 4) $\lambda = 7, \mu = -5$

4. $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is
   1) 0 2) 1 3) –1 4) i

5. If z is a complex number such that $z \in \mathbb{C}\setminus\mathbb{R}$ and $z + \frac{1}{z} \in \mathbb{R}$, then |z| is
   1) 0 2) 1 3) 2 4) 3

6. If $z = x+iy$ is a complex number such that $|z+2| = |z-2|$, then the locus of z is
   1) real axis 2) imaginary axis 3) ellipse 4) circle

7. A zero of $x^3 + 64$ is
   1) 0 2) 4 3) $4i$ 4) –4

8. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is
   1) 2 2) 4 3) 1 4) ∞

9. $\text{Sin}^{-1}\frac{3}{5} - \cos^{-1}\frac{12}{13} + \text{sec}^{-1}\frac{5}{3} - \text{cosec}^{-1}\frac{13}{12}$ is equal to
   1) 2π 2) π 3) 0 4) $\tan^{-1}\frac{12}{65}$

10. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to
    1) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ 2) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ 3) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ 4) $\tan^{-1}\left(\frac{1}{2}\right)$

11. If $|x| \leq 1$, then $2\tan^{-1}x - \sin^{-1}\frac{2x}{1+x^2}$ is equal to
    1) $\tan^{-1}x$ 2) $\sin^{-1}x$ 3) 0 4) π

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12. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the Distancebetween the foci is

1) $\frac{4}{3}$ 2) $\frac{4}{\sqrt{3}}$ 3) $\frac{2}{\sqrt{3}}$ 4) $\frac{3}{2}$

13. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ having centre at (0, 3) is

1) $x^2+y^2-6y-7=0$ 2) $x^2+y^2-6y+7=0$

3) $x^2+y^2-6y-5=0$ 4) $x^2+y^2-6y+5=0$

14. The values of m for which the line $y=mx+2\sqrt{5}$ touches the hyperbola $16x^2-9y^2=144$ are the roots of $x^2-(a+b)x-4=0$, then the value of $(a+b)$ is

1) 2 2) 4 3) 0 4) -2

15. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+\hat{j}$, $\vec{c}=\hat{i}$ and $(\vec{a}\times\vec{b})\times\vec{c}=\lambda\vec{a}+\mu\vec{b}$, then the value of $\lambda+\mu$ is

1) 0 2) 1 3) 6 4) 3

16. The angle between the lines $\frac{x-2}{3}=\frac{y+1}{-2}, z=2$ and $\frac{x-1}{1}=\frac{2y+3}{3}=\frac{z+5}{2}$ is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

17. If the direction cosines of a line are $\frac{1}{c},\frac{1}{c},\frac{1}{c}$ then

1) $c=\pm 3$ 2) $c=\pm\sqrt{3}$ 3) $c>0$ 4) $0<c<1$

18. Domain of $\text{cosec}^{-1}(x)$ *is*

1) $R\setminus(-1,1)$ 2) $R\setminus\{-1,1\}$ 3) $[-\frac{\pi}{2},\frac{\pi}{2}]$ 4) $R-\{0\}$

19. The eccentricity of an ellipse $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1, a<b$ is

1) $e=\sqrt{1-\frac{b^2}{a^2}}<1$ 2) $e=\sqrt{1-\frac{b^2}{a^2}}>1$

3) $e=\sqrt{1-\frac{a^2}{b^2}}<1$ 4) $e=\sqrt{1+\frac{a^2}{b^2}}<1$

20. Which of the following is not equivalent to $\vec{a}.(\vec{b}\times\vec{c})$

1) $-\vec{a}.(\vec{c}\times\vec{b})$ 2) $\vec{c}.(\vec{b}\times\vec{a})$ 3) $-\vec{b}.(\vec{c}\times\vec{a})$ 4) $(\vec{c}\times\vec{a}).\vec{b}$

PART – II

II. Answer any seven questions. Question No.30 is compulsory. **7 x 2 = 14**

21. If adj $A=\begin{bmatrix}-1 & 2 & 2\\ 1 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, find A^{-1}.

22. Find the rank of the matrix $\begin{bmatrix}1 & 2 & 3\\ 2 & 1 & 4\\ 3 & 0 & 5\end{bmatrix}$ by reducing it to a row-echelon form.

23. Show that $|3z-5+i|=4$ represents a circle, andfind its centre and radius.

24. Prove that z is real if and only if $z=\bar{z}$

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25. If α, β and γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \frac{1}{\beta\gamma}$ in terms of the coefficients.

26. Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ (in radians and degrees).

27. Find centre and radius of the circles: $2x^2 + 2y^2 - 6x + 4y + 2 = 0$

28. Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$

29. If $2\hat{i} - \hat{j} + 3\hat{k}$, $3\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + m\hat{j} + 4\hat{k}$ are coplanar, find the value of m.

30. If the planes $\vec{r}.(2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r}.(4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel then find the values of λ, μ.

PART – III

III. Answer any Seven questions. Question No.40 is compulsory. **7 x 3 = 21**

31. Verify the property $(A^T)^{-1} = (A^{-1})^T$ with $A = \begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$

32. Solve the system of linear equations by matrix inversion method: $2x - y = 8$, $3x + 2y = -2$.

33. If $|z| = 2$, show that $8 \leq |z + 6 + 8i| \leq 12$.

34. Show that the polynomial equation $9x^9 + 2x^5 - x^4 - 7x^2 + 2 = 0$ has at least six imaginary roots.

35. Show that $\cot(\sin^{-1}x) = \frac{\sqrt{1-x^2}}{x}, -1 \leq x \leq 1$ and $x \neq 0$

36. Find the value of $2\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right)$

37. If the normal at the point 't_1' on the parabola $y^2 = 4ax$ meets the parabola again at the point 't_2', then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

38. Find the magnitude and the direction cosines of the torque about the point (2, 0, –1) of a force $2\hat{i} + \hat{j} - \hat{k}$ whose line of action passes through the origin.

39. Find the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.

40. Find the real numbers x and y if $(x - iy)(3 + 5i)$ is conjugate of $-6 - 24i$.

41. Find the real numbers x and y if $(x-iy)(3+5i)$ is conjugate of -6 - 24i .

PART – IV

IV. Answer all the questions. **7 x 5 = 35**

41. (*a*) Solve the systems of linear equations by Cramer's rule:

$3x + 3y - z = 11$, $2x - y + 2z = 9$, $4x + 3y + 2z = 25$.

(OR)

(*b*) An amount of Rs.65,000 is invested in three bonds at the rates of 6%, 8% and 9% per annum respectively. The total annual income is Rs.4,800. The income from the third bond is Rs.600 more than that from the second bond. Determine the price of each bond.
(Use Gaussian elimination method.)

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42. (*a*)Find the value of $\left(\frac{1+\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}}{1+\sin\frac{\pi}{10}-i\cos\frac{\pi}{10}}\right)^{10}$

(OR)

(*b*) If $z=x+iy$ and $\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{2}$, then show that $x^2+y^2=1$.

43. (*a*)If the equations $x^2+px+q=0$ and $x^2+p'x+q'=0$ have a common root, show that it must be equal to $\frac{pq'-p'q}{q-q'}$ or $\frac{q-q'}{p'-p}$.

(OR)

(*b*)If $2+i$ and $3-\sqrt{2}$ are roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$, find all roots.

44. (*a*)Find the number of solution of the equation $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}(3x)$

(OR)

(*b*)If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ and $0<x, y, z<1$, show that $x^2+y^2+z^2+2xyz=1$.

45. (*a*)Find the vertex, focus, directrix, and length of the latus rectum of the parabola $x^2-4x-5y-1=0$.

(OR)

(*b*)A semielliptical archway over a one-way road has a height of $3m$ and a width of $12m$. The truck has a width of $3m$ and a height of $2.7\ m$. Will the truck clear the opening of the archway?

46. (*a*)Show that the line x+y= 1 is a tangent to the hyperbola $2x^2-3y^2=6$. Also find the coordinates of the point of contact.

(OR)

(*b*)Find the parametric form of vector equation of the straight line passing through (–1, 2, 1) and parallel to the straight line $\vec{r}=(2\hat{i}+3\hat{j}-\hat{k})+t(\hat{i}-2\hat{j}+\hat{k})$ and hence find the shortest distance between the lines.

47. (*a*) Find the vector parametric, vector non-parametric and Cartesian form of the equation of the plane passing through the points (–1, 2, 0), (2, 2, –1) and parallel to the straight line $\frac{x-1}{1}=\frac{2y+1}{2}=\frac{z+1}{-1}$.

(OR)

(*b*) Prove by vector method that $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$.

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